

BIOS vs UEFI

A Comparison of Legacy and Modern PC Firmware

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Comparison Table

Feature	BIOS	UEFI
Definition	This stands for Basic Input Output System. It is firmware stored on a chip on the motherboard, and its job is to wake up the hardware and hand control over to the operating system.	This stands for Unified Extensible Firmware Interface. It is the modern firmware standard that was built to take over from BIOS, and it comes with a much fuller pre boot environment.
Boot Process	It reads the Master Boot Record in an old 16 bit mode, runs the Power On Self Test one step at a time, then loads the bootloader.	It can run in 32 bit or 64 bit mode, works with both GPT and MBR, and can start up faster thanks to Fast Boot and parallel hardware checks.
Maximum Disk Support	Limited to drives of about 2.2 TB because of how the MBR partition scheme works.	Can support drives well beyond 2.2 TB, in theory up to 9.4 ZB, because it uses GPT.
Partition Style	Uses the Master Boot Record, which only allows up to 4 primary partitions.	Uses the GUID Partition Table, which allows up to 128 partitions on a single disk.
Security Features	Has very little built in protection, and it does not check for malware or rootkits during startup.	Comes with Secure Boot, which checks the digital signature of the OS loader before it runs, and it also works with TPM hardware.
Speed	Tends to boot slower because it checks hardware one piece at a time using older 16 bit code.	Boots noticeably faster because it can check hardware in parallel and supports Fast Boot.
Advantages	Simple, lightweight, and works on almost any older hardware or operating system.	Faster startup, support for large drives and more partitions, Secure Boot, a mouse friendly menu, and even network tools for troubleshooting.
Disadvantages	Slow to boot, capped at 2.2 TB, offers no real security, and only has a plain text interface.	A bit more complex to configure, which can trip up beginners, and it can occasionally run into issues with very old hardware.
Modern Usage	Mostly found on old or legacy machines at this point.	The standard on nearly every new computer sold today, and Windows 11 requires it.

Why UEFI Has Largely Replaced BIOS

For a long time, BIOS was the quiet gatekeeper of every PC. The moment you hit the power button, it was BIOS doing the unglamorous work of waking up the hardware, running a quick self check, and handing things off to the operating system. It did its job well, and for most of computing history that was good enough. But BIOS was built for a different era, back when hard drives were small, hardware was simple, and nobody was really worried about firmware level malware. As computers grew more powerful and storage grew much larger, BIOS started to show its age, and the cracks became harder to ignore.

The Storage Problem

The clearest limitation is storage. BIOS relies on the Master Boot Record partitioning scheme, which caps a bootable drive at roughly 2.2 TB and only allows four primary partitions. That ceiling made sense back when multi terabyte drives sounded like science fiction, but today even a budget laptop can ship with a drive that blows right past it. UEFI fixes this with the GUID Partition Table, which can support drives into the zettabyte range and allow up to 128 partitions, so storage stops being a bottleneck for the foreseeable future.

Speed and User Experience

BIOS boots using an old 16 bit real mode and checks hardware one piece at a time, which adds up to a noticeably slower startup. UEFI runs in 32 bit or 64 bit mode, can check devices in parallel, and supports features like Fast Boot that cut startup time down quite a bit. UEFI also brought something BIOS never really had, and that is an interface people can actually use. Instead of a plain blue and white text menu navigated entirely with the keyboard, UEFI gives you graphical menus, mouse support, and sometimes even network access for remote troubleshooting. It sounds like a small thing, but it makes the firmware feel like it belongs in the same decade as the rest of the computer.

Security Was the Real Turning Point

The biggest reason UEFI won out is probably security. BIOS has no built in way to check whether the code it is about to run is legitimate, and that gap leaves room for bootkits and rootkits to take over a system before the operating system, or any antivirus software, even gets a chance to load. UEFI closes that gap with Secure Boot, which checks the digital signature of the bootloader before letting it run, and it also pairs naturally with TPM hardware for extra protection. As more threats start targeting the boot process itself, this one feature has become close to essential on any modern system.

Industry Wide Momentum

None of this happened on its own. Motherboard makers, chipmakers, and operating system companies all moved in the same direction at roughly the same time. Microsoft made Secure Boot a requirement for Windows 8 certification and later made UEFI itself a hard requirement for Windows 11, which pretty much ended the production of new BIOS only hardware. Combined with everything UEFI does better on the technical side, that industry wide push turned the switch from something optional into something that was really only a matter of time.

The Bottom Line

BIOS was not a failure. It powered personal computers for more than thirty years and did exactly what it was built to do. It was just built for a smaller and far less hostile computing world. UEFI answers almost every one of BIOS's weak points, from bigger drives and faster boots to a friendlier interface and, most importantly, real protection against attacks that target the boot process. That combination is why nearly every computer built in the last decade runs on UEFI, and why BIOS mostly survives now in old machines and in the memories of the technicians who used to configure it.